

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	46.0 / Female
Department	Gynaecology
Diagnosis	Acute cystitis
UTI Classification	Uncomplicated UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Dysuria;Frequency

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9100.0	cells/ μ L	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	10.0	mg/L	< 10
Rft Serum Creatinine	0.9	mg/dL	0.6 - 1.3
Serum Uric Acid	4.3	mg/dL	3.5 - 7.2
Blood Urea	26.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida albicans
Bacteria Type	Fungus (Yeast; the most common cause of candiduria)

Antibiotic Susceptibility Profile

Predicted Sensitive	Amphotericin B, Voriconazole
Predicted Resistant	Fluconazole

Recommended Therapy

Liposomal Amphotericin B

Dosage: 5 mg/kg IV once daily (e.g., 250 mg for a 50 kg patient)

Precautions: Monitor serum creatinine and electrolytes daily; pre-hydrate with 1 L normal saline before infusion; avoid concomitant nephrotoxic agents (e.g., aminoglycosides, NSAIDs). No significant hepatic metabolism, but monitor for infusion reactions.

Rationale: Liposomal formulation reduces nephrotoxicity while retaining potent activity against Candida albicans, making it a reliable choice for candiduria.

Voriconazole

Dosage: Loading dose 6 mg/kg IV q12h for 2 doses, then maintenance 4 mg/kg IV q12h (≈200 mg PO BID after initial 2 days).

Precautions: Check baseline hepatic panel; avoid in patients with severe liver disease. Monitor for visual disturbances, photosensitivity, and drug interactions (CYP2C19 inhibitors). Adjust dose if renal function declines but not required for standard dosing.

Rationale: Voriconazole offers broad antifungal coverage including Candida albicans and has good urinary penetration, making it an effective alternative when amphotericin B is contraindicated or not tolerated.

Antibiotic Drug History

Clinical Summary

Clinical Summary - 46-year-old female with uncomplicated cystitis

- **Infection:** Acute uncomplicated cystitis (bladder) caused by *Candida albicans* (most common fungal uropathogen).
- **Resistance profile:** Fluconazole-resistant.
- **Sensitivity profile:** Potently active against amphotericin B and voriconazole.

Recommended antifungal therapy

1. Liposomal Amphotericin B

- **Dose:** 5 mg/kg IV once daily (e.g., 250 mg for a 50 kg patient).
- **Rationale:** Liposomal formulation offers high urinary concentrations with markedly reduced nephrotoxicity, making it a first-line choice for candiduria.
- **Precautions:** Daily monitoring of serum creatinine and electrolytes; pre-hydrate with 1 L normal saline; avoid concurrent nephrotoxins (aminoglycosides, NSAIDs); watch for infusion reactions.

2. Voriconazole (alternative if amphotericin B is contraindicated or not tolerated)

- **Dose:** Loading 6 mg/kg IV q12h for 2 doses, then 4 mg/kg IV q12h (≈200 mg PO BID after 2 days).
- **Rationale:** Broad antifungal activity, excellent urinary penetration, and a favorable safety profile in most patients.
- **Precautions:** Baseline hepatic panel; avoid in severe liver disease; monitor for visual disturbances, photosensitivity, and drug interactions (CYP2C19 inhibitors). Dose adjustment not required for standard renal function, but watch for renal decline.

Key considerations

- Prior antibiotics (fosfomycin, azithromycin) are ineffective against yeast; switch to antifungal therapy is essential.
- Monitor renal function closely, especially with amphotericin B.
- Ensure adequate hydration and avoid nephrotoxic agents.
- Educate the patient on signs of infusion reactions and visual changes with voriconazole.

This concise plan targets the resistant organism, balances efficacy with safety, and provides clear monitoring parameters for optimal patient outcomes.